Katherine Titterton

Imaging, signal & ML scientist-engineer · neuroscience

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Neuroscientist-engineer at the seam of imaging, signal extraction, and machine learning. Eight years building the instruments, datasets and models that turn biological signal into measurement, from voltage-clamp recordings and expansion microscopy to PyTorch segmentation over 100,000+ compound image sets.

EXPERIENCE

[n] = publication number on page 2

Axiom Bio · Founding Member of the Technical Staff — San Francisco, CA — 2023–present

- Led image labeling and PyTorch U-Net segmentation of 100,000+ compound images, in production 1+ year.
- Invented two image-based bioactivity-detection methods ~10× more sensitive than conventional assays.
- Oversaw assay development and QC for two of the largest Cell Painting datasets; shaped the product.
  Nat Commun 2026 [4] · Cell Syst 2026 [3]

Insitro · Associate Scientist III, Arrayed Screening — South San Francisco, CA — 2022–2023

- Onboarded tuberous sclerosis and ALS screens, from assay development to automated 384-well imaging.
- Trained two screen operators; analyzed hundreds of plates in Python.
- Developed a label-free phase-contrast timecourse imaging method to track live neuron growth.

MIT · Technical Associate II, Synthetic Neurobiology Group — Cambridge, MA — 2021–2022

- Imaged 30+ proteins at nanoscale by optimizing expansion revealing (mExR) for 7+ staining rounds on brain tissue.
- Built the ImageJ/Fiji and MATLAB analysis pipelines behind the published method.
  Nat Commun 2024 [6]

AbbVie · Associate Scientist II, Foundational Neuroscience Center — Cambridge, MA — 2018–2021

- Established tau-aggregation screens in primary and iPSC neurons, from assay development to R lead ranking.
- Designed microfluidic neuron cultures that model tau spread between cells, for testing propagation inhibitors.
  Mol Neurodegener Adv 2026 [1] · iScience 2026 [2] · J Biol Chem 2025 [5] · iScience 2022 [7]

Massachusetts General Hospital · Research Technician I — Boston, MA — 2017–2018

- Recorded voltage-clamp currents from GABA-A receptors in *Xenopus* oocytes to study how anesthetics act.
  Eur J Med Chem 2019 [8]

PROJECTS

OpenTronsformers · LLM copilot for the OT-2 lab robot; 2nd place, BioHack NYC — 2023

Open source · Diffusion video pipeline and an agent-skills library, github.com/kat-titter — 2026

EDUCATION

Colgate University · B.A. Cellular Neuroscience, *cum laude* — 2013–2017
Minor in Mathematical Systems Biology. Co-captain, Division I Track & Field.

Harvard Extension School · Graduate coursework in Python and computer vision — 2017–2020

SKILLS

ML & data: PyTorch, U-Net segmentation (SAM evaluated), image embeddings and clustering, Python (NumPy, SciPy, scikit-learn, pandas), R, statistics, Claude Code, Cursor

Imaging & signal: high-content and confocal microscopy (Opera Phenix, Operetta), expansion microscopy, phase-contrast timecourse, voltage-clamp electrophysiology, ImageJ/Fiji, MATLAB

Lab & automation: primary and iPSC neuron culture, immunostaining, assay development, high-throughput screening, microfluidics, liquid handlers and schedulers (OpenTrons, BioTek, Echo, HighRes)

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PEER-REVIEWED PUBLICATIONS

8 papers · newest first

[1] Yanamandra K, …, **Titterton K**, …, Langlois X. Conformation-specific anti-pathological tau antibody raised against AD brain-derived seed-competent tau reduces tau pathology in transgenic mice. *Mol Neurodegener Adv*. 2026;2(1):43. doi:10.1186/s44477-026-00049-6 ABBVIE

[2] Sebastian RM, …, **Titterton K**, …, Wu JW. SynCAR platform for capturing neuronal synaptic proteopathic seeds. *iScience*. 2026;29(7):116362. doi:10.1016/j.isci.2026.116362 ABBVIE

[3] Ewald JD, **Titterton KL**, …, Singh S. Cell Painting for cytotoxicity and mode-of-action analysis in primary human hepatocytes. *Cell Syst*. 2026;17(5):101566. doi:10.1016/j.cels.2026.101566 AXIOM BIO

[4] Seal S, …, **Titterton K**, …, Carpenter AE. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nat Commun*. 2026;17(1). doi:10.1038/s41467-026-68725-5 AXIOM BIO featured on axi.om

[5] Wu JW, **Titterton K**, …, Langlois X. Development of a robust, physiologically relevant neuronal tau aggregation assay for tau-related drug discovery. *J Biol Chem*. 2025;301(12):110853. doi:10.1016/j.jbc.2025.110853 ABBVIE

[6] Kang J, …, **Titterton K**, …, Boyden ES. Multiplexed expansion revealing for imaging multiprotein nanostructures in healthy and diseased brain. *Nat Commun*. 2024;15(1):9722. doi:10.1038/s41467-024-53729-w MIT

[7] Manos JD, …, **Titterton K**, …, Langlois X. Uncovering specificity of endogenous TAU aggregation in a human iPSC-neuron TAU seeding model. *iScience*. 2022;25(1):103658. doi:10.1016/j.isci.2021.103658 ABBVIE

[8] Wu B, …, **Titterton K**, …, Bruzik KS. Inhibitable photolabeling by neurosteroid diazirine analog in the β3-subunit of human heteropentameric type A GABA receptors. *Eur J Med Chem*. 2019;162:810-824. doi:10.1016/j.ejmech.2018.11.019 MGH

SELECTED CONFERENCE PRESENTATIONS & POSTERS

[1] **Titterton K**, Boiko D, Cabrera ÁA, Bäuerle A, Tyagi A, Saneinejad S, Beatson A, White B. Accurate and interpretable in silico clinical risk assessment for drug-induced liver injury (DILI) from molecular structure. *Society of Toxicology Annual Meeting*, 2025. poster

[2] **Titterton K**, et al. Development of a tau propagation neuronal microfluidic cell culture model for assessing inhibitors of tau propagation. *Society for Neuroscience Global Connectome*, 2021. poster

[3] **Titterton K**, et al. High content analysis of tau aggregation in primary neurons: screening anti-tau antibodies and modeling tauopathy progression in vitro. *SBI2 High Content Meeting*, Harvard Medical School, 2019. poster